

Literature Status: Order-Reversing Operators on Convex Functions

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Status

This is a `literature_already_answered` packet for the self-map branch of the open problem in Iusem–Reem–Svaiter, arXiv:1212.1120. It is not a new solution produced by this run.

Source Question

In Section 3 of arXiv:1212.1120, after characterizing fully order reversing maps

$$S : \mathcal{C}(X) \rightarrow \mathcal{C}_{w^*}(X^*),$$

Iusem–Reem–Svaiter write that the characterization of fully order reversing operators

$$S : \mathcal{C}(X) \rightarrow \mathcal{C}(X)$$

and, in the nonreflexive case, $S : \mathcal{C}(X) \rightarrow \mathcal{C}(X^*)$, is left as an open problem. Here $\mathcal{C}(X)$ denotes the proper lower semicontinuous convex functions on the Banach space X .

Answering Paper

Cheng–Luo, *On order preserving and order reversing mappings defined on cones of convex functions*, arXiv:1708.06548, explicitly quote this Iusem–Reem–Svaiter question in their introduction. Their Theorem 1.5 states that a fully order reversing self-map

$$S : \text{conv}(X) \rightarrow \text{conv}(X)$$

exists if and only if X is reflexive and linearly isomorphic to its dual X^* . Their Theorem 1.6 gives the normal form: for every such S , there are a linear isomorphism $U : X \rightarrow X^*$, vectors $x_0^*, \varphi_0 \in X^*$, a scalar $\alpha > 0$, and $r_0 \in \mathbb{R}$ such that

$$(Sf)(x) = \alpha (\mathcal{F}f)(Ux + x_0^*) + \langle \varphi_0, x \rangle + r_0, \quad x \in X,$$

where $\mathcal{F}$ is the Fenchel transform.

Scope

The later paper explicitly knows it is addressing the Iusem–Reem–Svaiter open problem: it cites the source, quotes the problem passage, and says that the resulting theorems resolve the self-map existence and representation questions. This packet records that identification so the run does not treat the self-map branch of arXiv:1212.1120 as a fresh open target.

References

- A. N. Iusem, D. Reem, and B. F. Svaiter, *Order preserving and order reversing operators on the class of convex functions in Banach spaces*, arXiv:1212.1120; J. Funct. Anal. 268 (2015), 73–92.
- L. Cheng and S. Luo, *On order preserving and order reversing mappings defined on cones of convex functions*, arXiv:1708.06548.